GATE ME 90 Days Challenge

Day 37/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Thermodynamics	**First Law**	**68 days to GATE**

How to use this set: 15 questions (7 MCQ, 8 NAT) on the first law for closed systems (cycles, processes, sign convention, piston–cylinder devices) and for control volumes: the steady flow energy equation for turbines, compressors, pumps, nozzles, diffusers, throttling valves, heat exchangers and mixing chambers, and the filling of tanks. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Which of the following is **NOT** a thermodynamic property of a system?

(A) Internal energy U

(B) Enthalpy H = U + pV

(C) The integral ∫p dV between two states

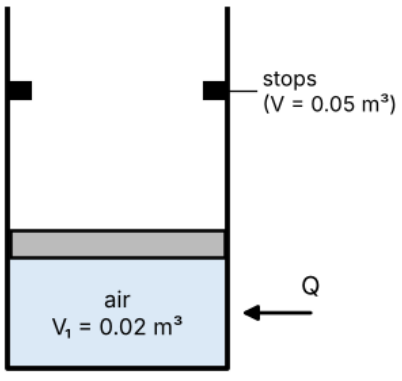
(D) Specific volume v

Q2 NAT

A closed system contains 2 kg of gas. During a compression process 120 kJ of work is done on the gas and 50 kJ of heat is rejected. The change in specific internal energy (in kJ/kg) is _____

Q3 NAT

Air (R = 0.287, c_V = 0.718 kJ/kg·K) at 300 kPa and 300 K occupies 0.02 m³ under a frictionless piston, as shown. Heat is added until the piston just reaches the stops (V = 0.05 m³), and then further heat is added until the pressure is 600 kPa. The total heat supplied (in kJ, round off to one decimal place) is _____



Q4 MCQ

An ideal gas is compressed slowly and isothermally in a closed system. Which statement is correct?

(A) $\Delta U > 0$ because work is done on the gas

(B) $\Delta U = 0$, and the heat rejected equals the work done on the gas

(C) $Q = 0$ because the temperature does not change

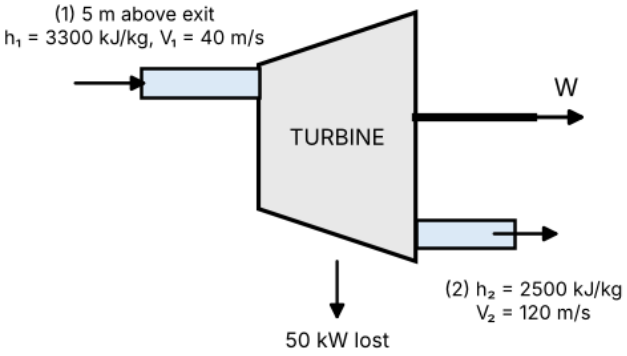
(D) $\Delta U = W$ because there is no heat transfer

Q5 NAT

1 kg of air ($R = 0.287$ kJ/kg·K) expands reversibly and isothermally at 400 K from 500 kPa to 100 kPa. The heat supplied (in kJ, round off to one decimal place) is _____

Q6 NAT

Steam flows steadily through the turbine shown at 5 kg/s. The inlet is 5 m above the exit. Taking $g = 9.81$ m/s², the power output (in kW, round off to the nearest integer) is _____



Q7 MCQ

Steam at negligible velocity enters an adiabatic nozzle at 2 kg/s. Its enthalpy falls by 300 kJ/kg, and the specific volume at exit is 0.8 m³/kg. The exit area of the nozzle is about

(A) 14.6 cm²

(B) 41.3 cm²

(C) 29.2 cm²

(D) 20.7 cm²

Q8 NAT

Air ($c_p = 1.005$ kJ/kg·K) enters a steady-flow compressor at 1.2 kg/s and 290 K. The power input is 180 kW and 12 kW of heat is lost from the casing. Neglecting kinetic and potential energy changes, the exit temperature (in K, round off to one decimal place) is _____

Q9 MCQ

In a condenser, 0.5 kg/s of steam enters with $h = 2400$ kJ/kg and leaves as condensate with $h = 190$ kJ/kg. Cooling water ($c_p = 4.18$ kJ/kg·K) rises in temperature by 10 °C. The required cooling-water flow rate is about

(A) 26.4 kg/s

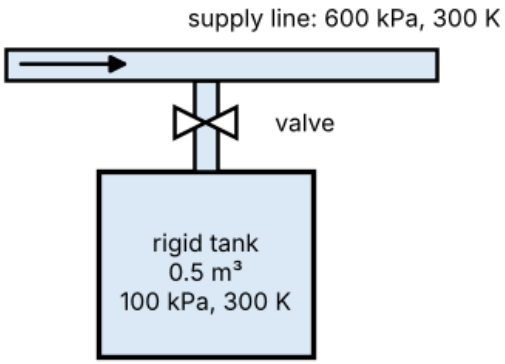
(B) 52.9 kg/s

(C) 28.7 kg/s

(D) 2.6 kg/s

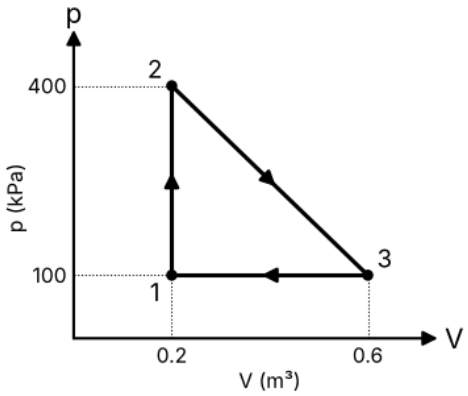
Q10 NAT

A rigid insulated tank of 0.5 m³ contains air at 100 kPa and 300 K. It is connected through a valve to a supply line carrying air at 600 kPa and 300 K, as shown. The valve is opened and closed when the tank pressure reaches 600 kPa. Taking c_p = 1.005 and c_v = 0.718 kJ/kg·K (ideal gas), the final temperature in the tank (in K, round off to one decimal place) is _____



Q11 MCQ

A closed system executes the cycle 1–2–3–1 shown on the p–V diagram. The net heat transfer during the cycle is



(A) 60 kJ rejected
(B) 60 kJ supplied
(C) 120 kJ supplied
(D) 30 kJ rejected

Q12 MCQ

Which statement follows from the first law of thermodynamics?

(A) No engine working in a cycle can convert all the heat supplied into work
(B) Heat cannot flow by itself from a colder body to a hotter body
(C) A perpetual motion machine of the first kind (producing work continuously without any energy input) is impossible
(D) Two bodies in thermal equilibrium with a third are in thermal equilibrium with each other

Q13 NAT

2 kg of wet steam (dryness 0.4) at 200 kPa is heated at constant pressure in a piston–cylinder device until it reaches 200 °C. At 200 kPa: h_f = 504.7 kJ/kg, h_{fg} = 2201.6 kJ/kg; at 200 kPa, 200 °C: h = 2870.5 kJ/kg. The heat supplied (in kJ, round off to the nearest integer) is _____

Q14 MCQ

In a reversible steady-flow pump, water (v = 0.001 m^3/kg, incompressible) is raised from 100 kPa to 5 MPa. Neglecting kinetic and potential energy changes, the work input per kg is

(A) 5.0 kJ/kg
(B) 0.49 kJ/kg
(C) 49 kJ/kg
(D) 4.9 kJ/kg

Q15 NAT

Two air streams mix in a steady-flow chamber: 2 kg/s at 400 K and 3 kg/s at 300 K. The chamber loses 20 kW of heat. With c_p = 1.005 kJ/kg·K and negligible KE and PE changes, the exit temperature (in K, round off to one decimal place) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(C) The integral ∫p dV between two states
Q2	NAT	35
Q3	NAT	69.0 (accept 68.5 to 69.5)
Q4	MCQ	(B) ΔU = 0, and the heat rejected equals the work done on the gas
Q5	NAT	184.8 (accept 183.5 to 186.0)
Q6	NAT	3918 (accept 3900 to 3936)
Q7	MCQ	(D) 20.7 cm²
Q8	NAT	429.3 (accept 427.5 to 431.0)
Q9	MCQ	(A) 26.4 kg/s
Q10	NAT	393.7 (accept 391.7 to 395.7)
Q11	MCQ	(B) 60 kJ supplied
Q12	MCQ	(C) A perpetual motion machine of the first kind (producing work continuously without any energy input) is impossible
Q13	NAT	2970 (accept 2955 to 2985)
Q14	MCQ	(D) 4.9 kJ/kg
Q15	NAT	336.0 (accept 334.5 to 337.5)

Detailed Solutions

Q1 **Answer: (C) The integral ∫p dV between two states**

Concept: A property depends only on the state; its change between two states is independent of the path.

U, H and v are point functions. ∫p dV is the boundary work, which is the area under the process curve on the p–V diagram, so it changes with the path.

δQ and δW are inexact differentials; dU = δQ − δW is exact.

Exam tip: If a quantity's cyclic integral is zero for every cycle, it is a property (that is how the first law proves U is a property).

Q2 **Answer: 35**

Concept: First law for a closed system: Q = ΔU + W, with W positive when done by the system and Q positive when supplied.

Q = −50 kJ, W = −120 kJ

ΔU = Q − W = −50 − (−120) = +70 kJ

Δu = 70/2 = **35 kJ/kg**

Exam tip: Write the signs first and then substitute; most errors in first-law questions are sign errors.

Q3 **Answer: 69.0 (accept 68.5 to 69.5)**

Concept: Process 1–2 at constant pressure (piston free) gives boundary work; process 2–3 at constant volume (piston on stops) gives no work. Then Q = ΔU + W.

$m = pV/RT = 300 \times 0.02/(0.287 \times 300) = 0.06969$ kg

$W = p(V_2 - V_1) = 300 \times (0.05 - 0.02) = 9$ kJ (no work in 2–3)

$T_3 = p_3V_3/(mR) = 600 \times 0.05/(0.06969 \times 0.287) = 1500$ K

$\Delta U = m\, c_V(T_3 - T_1) = 0.06969 \times 0.718 \times 1200 = 60.04$ kJ

$Q = 60.04 + 9 =$ **69.0 kJ**

Exam tip: Draw the path on the p–V diagram: a horizontal line then a vertical line. Work is only the area under the horizontal part.

Q4 **Answer: (B) ΔU = 0, and the heat rejected equals the work done on the gas**

Concept: For an ideal gas, U depends only on T.

Isothermal → ΔU = 0, so Q = W. In compression W is negative, so Q is equally negative: all the work done on the gas leaves as heat.

Option (C) confuses temperature change with heat transfer: an isothermal process is not adiabatic.

Exam tip: "No temperature change" does not mean "no heat". Heat and temperature change are different things.

Q5 **Answer: 184.8 (accept 183.5 to 186.0)**

Concept: Isothermal ideal gas: ΔU = 0, so Q = W = $mRT \ln(p_1/p_2)$.

$Q = 1 \times 0.287 \times 400 \times \ln(500/100) = 114.8 \times 1.6094$

Q = **184.8 kJ**

Exam tip: For isothermal work use $\ln(V_2/V_1) = \ln(p_1/p_2)$; note the pressure ratio is inverted.

Q6 **Answer: 3918 (accept 3900 to 3936)**

Concept: Steady flow energy equation: $\dot{W} = \dot{m}[(h_1 - h_2) + (V_1^2 - V_2^2)/2 + g(z_1 - z_2)] + \dot{Q}$, with $\dot{Q} = -50$ kW (heat lost).

$\Delta h = 3300 - 2500 = 800$ kJ/kg

$\Delta KE = (40^2 - 120^2)/2000 = -6.4$ kJ/kg; $\Delta PE = 9.81 \times 5/1000 = 0.049$ kJ/kg

$\dot{W} = 5 \times (800 - 6.4 + 0.049) - 50 = 3968.2 - 50 =$ **3918 kW**

Exam tip: PE terms are usually tiny (1 kJ/kg ≈ 102 m of height). Always divide V^2 and gz by 1000 to get kJ/kg.

Q7 **Answer: (D) 20.7 cm²**

Concept: Nozzle energy balance gives the exit velocity; continuity $\dot{m} = AV/v$ gives the area.

$V_2 = \sqrt{(2 \times 300\,000)} = 774.6$ m/s

$A_2 = \dot{m}v_2/V_2 = 2 \times 0.8/774.6 = 2.066 \times 10^{-3}$ m² = **20.7 cm²**

(C) comes from forgetting the 2 in $\sqrt{(2\Delta h)}$; (B) doubles the mass flow.

Exam tip: A quick check: $V = 44.72\sqrt{(\Delta h \text{ in kJ/kg})}$ m/s for negligible inlet velocity.

Q8 **Answer: 429.3 (accept 427.5 to 431.0)**

Concept: SFEE for a compressor: $\dot{m}\, c_p(T_2 - T_1) = \dot{W}_{in} - \dot{Q}_{loss}$.

$\dot{m}\, c_p \Delta T = 180 - 12 = 168$ kW

$\Delta T = 168/(1.2 \times 1.005) = 139.3$ K

$T_2 = 290 + 139.3 =$ **429.3 K**

Exam tip: Heat lost from a compressor reduces the exit enthalpy; the same power without heat loss would give 439.3 K.

Q9 **Answer: (A) 26.4 kg/s**

Concept: In a heat exchanger with no work and no heat loss, the enthalpy given up by one stream equals the enthalpy gained by the other.

Heat rejected by steam $= 0.5 \times (2400 - 190) = 1105$ kW

$\dot{m}_w = 1105/(4.18 \times 10) =$ **26.4 kg/s**

Exam tip: Condensers need roughly 50 kg of water per kg of steam: the steam's latent heat is very large.

Q10 **Answer: 393.7 (accept 391.7 to 395.7)**

Concept: Uniform-state filling: $m_2u_2 - m_1u_1 = (m_2 - m_1)h_{line}$ (no heat, no work).

With $m = pV/RT$: $c_v(p_2 - p_1)V/R = c_pT_L(p_2/T_2 - p_1/T_1)V/R$

$0.718 \times 500 = 1.005 \times 300 \times (600/T_2 - 100/300) \rightarrow 600/T_2 = 1.1907 + 0.3333 = 1.5240$

$T_2 = 600/1.5240 =$ **393.7 K** ($m_1 = 0.581$ kg, $m_2 = 2.655$ kg)

Exam tip: An evacuated tank would reach $\gamma T_{line} = 420$ K; the gas already present lowers the final temperature.

Q11 **Answer: (B) 60 kJ supplied**

Concept: For a cycle $\oint \delta Q = \oint \delta W$, and the net work is the area enclosed on the p–V diagram.

Area of the triangle $= \frac{1}{2} \times (0.6 - 0.2) \times (400 - 100) = 60$ kJ

The cycle runs clockwise (1→2 up, 2→3 down-right, 3→1 left), so the net work is done by the system and net heat is **60 kJ supplied**.

Exam tip: Clockwise on p–V = work output (engine); anticlockwise = work input (refrigerator/heat pump).

Q12 **Answer: (C) A perpetual motion machine of the first kind (producing work continuously without any energy input) is impossible**

Concept: The first law is the conservation of energy: in a cycle, net work output cannot exceed net heat input.

(A) is the Kelvin–Planck statement and (B) the Clausius statement of the second law; (D) is the zeroth law.

So a PMM1 is impossible: **(C)**.

Exam tip: PMM1 violates the first law; PMM2 (100 % heat-to-work in a cycle) violates the second law.

Q13 **Answer: 2970 (accept 2955 to 2985)**

Concept: Quasi-static constant-pressure process in a closed system: $Q = \Delta H = m(h_2 - h_1)$.

$h_1 = h_f + x\, h_{fg} = 504.7 + 0.4 \times 2201.6 = 1385.3$ kJ/kg

$Q = 2 \times (2870.5 - 1385.3) =$ **2970 kJ**

Exam tip: Using h instead of u at constant pressure accounts for the boundary work automatically.

Q14 **Answer: (D) 4.9 kJ/kg**

Concept: For a reversible steady-flow process with negligible ΔKE and ΔPE, $w = -\int v\, dp$ (work done by the fluid).

Work input $= v(p_2 - p_1) = 0.001 \times (5000 - 100) =$ **4.9 kJ/kg**

Contrast with closed-system work $\int p\, dV$, which is zero for an incompressible liquid.

Exam tip: Pump work is tiny compared with turbine work because v of liquid is about 1000 times smaller than v of steam.

Q15 **Answer: 336.0 (accept 334.5 to 337.5)**

Concept: Mixing chamber SFEE: $\Sigma \dot{m} h_{in} - \dot{Q}_{loss} = \Sigma \dot{m} h_{out}$.

Without heat loss: $T = (2 \times 400 + 3 \times 300)/5 = 340$ K

Heat loss lowers it by $20/(5 \times 1.005) = 3.98$ K

$T_3 = 340 - 3.98 =$ **336.0 K**

Exam tip: For the same c_p, adiabatic mixing gives the mass-weighted mean temperature.

What next?

Day 37 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 38/90: Thermodynamics, Second Law & Entropy. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.